

Task 2.2)

State:

Context - Traveller

State - TransportationMethod

ConcreteStateA - TeleportationCircle

ConcreteStateB - MerchantCaravan

ConcreteStateC - Horseback

We made the Traveller own the Transportation method to reduce the amount of memory owned by the main function to make freeing memory easier

Strategy:

Context - GameManager

Strategy - Route

ConcreteStrategyA - WarmestRoute

ConcreteStrategyB - CheapestRoute

ConcreteStrategyC - FastestRoute

Since the routefinding method can be changed on the fly, rather than storing the entire trip and generating it over, the Concrete Routes simply give the next Settlement according to the criteria.

Composite:

Tree - Place

BaseNode - Settlement

IntermediateNode - Region

We made the GameManager own the root of the world map to remove responsibility from main.

Decorator:

Component - Settlement

ConcreteComponent - City

Decorator - SettlementFeature

ConcreteDecoratorA - Food

ConcreteDecoratorB - Work

ConcreteDecoratorC - Lodging

ConcreteDecoratorD - Market

Rather than have the Place interface be the component, we made the BaseNode(Settlement) be the component so we can add interesting things for the player to experience within the cities, bringing them alive.

Abstract Factory:

AbstractFactory - BiomeFactory
ConcreteFactoryA - ForestFactory
ConcreteFactoryB - OceanFactory
AbstractProductA - Food
ConcreteProductA1 - OceanFood
ConcreteProductA2 - ForestFood
AbstractProductB - Market
ConcreteProductB1 - OceanMarket
ConcreteProductB2 - ForestMarket

We made the abstract factory produce concrete decorators which can be added to a city.

Task 2.3)

TeleportationCircle - State

Make it inherit from transportationMethod and implement the leave function then it slots in without needing to change anything else.

OceanFood - Decorator

It inherits from food, meaning it just needs to implement eat() then it can be used like any other decorator

WarmestRoute - Strategy

Inheriting from Route, as long as it implements getNextSettlement() it can be used in place of any pre-existing Route subclass.